AI for Redistricting: An Analysis of Rhode Island's State Senate Districts

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Code link: https://github.com/TianqiMin8/CS464FinalProject.git

1 Introduction

The goal of this report is to analyze Rhode Island's 2022 State Senate redistricting map for signs of partisan gerrymandering and to assess whether minority communities are adequately represented under the enacted plan. To do this, we use a computational ensemble approach: we generate thousands of alternative redistricting maps at random and compare the enacted map's partisan and compactness metrics against that distribution. A map that is a statistical outlier relative to the ensemble may be evidence of deliberate manipulation.

Redistricting is the process of redrawing the geographic boundaries of legislative districts. In the United States, this process occurs every ten years following the decennial census, as population shifts require districts to be redrawn to preserve the constitutional principle of "one person, one vote" [9]. Redistricting decisions are enormously consequential: the shape of a district determines which voters are grouped together and, in turn, which candidates are favored to win.

Gerrymandering occurs when district boundaries are drawn to advantage a particular party, group, or incumbent. It can take two broad forms: (1) *partisan gerrymandering*, in which maps are drawn to maximize the number of seats won by one political party, and (2) *racial gerrymandering*, in which maps either "pack" or "crack" communities of color to dilute their electoral influence. The Voting Rights Act of 1965 (VRA) is the primary federal tool for combating racial gerrymandering. Under Section 2 of the VRA, a redistricting plan is unlawful if it results in the denial or abridgment of the right to vote on account of race [10].

When a minority community is large, geographically concentrated, and votes cohesively, the VRA may require the creation of a majority-minority district (also called a *VRA district*), in which a district in which the minority group comprises a majority of the voting-age population, giving that community a realistic opportunity to elect its preferred candidate. A plaintiff can establish that a VRA district is required by satisfying the three *Gingles* criteria [10]:

1. The minority group is sufficiently large and geographically compact to form a majority in a single-member district (*Gingles 1*).

2. The minority group is politically cohesive, meaning its members tend to vote for the same candidates (*Gingles 2*).

3. The white majority votes sufficiently as a bloc to usually defeat the minority group's preferred candidate (*Gingles 3*).

All three criteria must be satisfied for a Section 2 claim to succeed. Rhode Island's growing Hispanic and Black populations, concentrated in Providence County, make VRA district analysis directly relevant to this report.

This report analyzes the Rhode Island State Senate district map adopted in 2022, following the 2020 decennial census. Rhode Island has 38 state senate seats. We use an ensemble of randomly generated alternative maps, constructed via a Markov Chain Monte Carlo (MCMC) process, to assess whether the enacted plan is an outlier in terms of partisan fairness and compactness.

Contextual Description of Rhode Island

When and by whom the map was created. Following the 2020 census, the Rhode Island General Assembly established an 18-member Special Commission on Reapportionment, composed of six members appointed by the Senate President, six by the House Speaker, and six from the general public [2]. The commission held its first meeting on September 9, 2021 and voted to approve new state legislative maps on January 12, 2022. The General Assembly passed the redistricting legislation (S 2162 / H 7323) on February 15, 2022, and Governor Dan McKee (D) signed it on February 16, 2022 [2]. The Senate approved the maps 29–9 and the House 57–6. It is notable that the enacted maps were not challenged in court.

Legal landscape. Rhode Island has a notable history of VRA litigation concerning its state senate map. In 2002, former state senator Harold Metts led a lawsuit alleging that the enacted senate plan violated Section 2 of the Voting Rights Act by failing to create sufficient majority-minority districts. The U.S. District Court initially dismissed the complaint, but on March 30, 2004 the U.S. Court of Appeals for the First Circuit reversed and remanded the case. While the case was pending on remand, the General Assembly enacted a revised senate plan (S 3137) on June 4, 2004, which effectively doubled minority representation in the Senate [11]. This litigation established an important precedent that Rhode Island's growing minority population must be meaningfully represented in the state senate.

A significant concern following the enactment of the 2022 maps was the state's failure to conduct or release a racial bloc voting analysis. Six civil rights organizations, Common Cause RI, the ACLU of Rhode Island, the Urban League of Rhode Island, the NAACP Providence Branch, ULMAC, and the National Coalition of 100 Black Women RI Chapter — sent a joint letter to the state's redistricting consultant demanding an explanation for his failure to use or release racial bloc voting analysis, despite being contractually required to do so [5]. The organizations further noted that despite Rhode Island's growing minority population since the last census, the newly approved House and Senate plans contained the *same number* of majority-minority districts as the maps drawn ten years prior. Racial bloc voting analysis serves as the primary tool used to assess Gingles 2 and Gingles 3, so its absence makes it impossible to determine whether the enacted map complies with the requirements of the Voting Rights Act.

A separate and ongoing issue has been prison-based gerrymandering. Rhode Island's Adult Correctional Institutions (ACI) are located in Cranston. For decades, incarcerated individuals were counted as residents of Cranston for redistricting purposes, inflating that district's apparent population and diminishing urban districts where many communities of color actually live. The ACLU of Rhode Island and Common Cause RI repeatedly urged the commission to address this practice [1]. In January 2022, the commission voted 15–1 to count individuals incarcerated for *less than two years* at their home addresses for redistricting purposes, which advocates described as a partial reform and characterized as "a small step" that did not fully eliminate the practice [6].

Partisan and racial landscape. Rhode Island is a strongly Democratic state. Democrats hold supermajorities in both chambers of the General Assembly; after the 2022 elections, they held 33 of 38 state senate seats [3]. In the 2020 Presidential election, Joe Biden carried Rhode Island by approximately 21 percentage points.

Racially, Rhode Island has grown more diverse over the past decade. According to the 2020 Census [12], the state population was approximately:

- 71.3% Non-Hispanic White
- 18.7% Hispanic or Latino (up from 12.4% in 2010 — a nearly 40% increase)
- 5.7% Non-Hispanic Black or African American
- 3.7% Non-Hispanic Asian

The Hispanic/Latino population is heavily concentrated in Providence County, where it represents approximately 24.3% of residents [4]. This geographic concentration is particularly relevant for Voting Rights Act analysis because it determines whether majority-minority districts are feasible under Gingles 1.

2 Methods

We analyze the Rhode Island State Senate map using an ensemble approach grounded in Markov Chain Monte Carlo (MCMC) sampling, implemented via the GerryChain library [7].

2.1 Data Collection and Preparation

All data were downloaded from the Redistricting Data Hub (RDH) Rhode Island page [8] in shapefile format. The RDH is the primary public repository for redistricting-ready Census and election data; it is the source specified in Lab 4 of this course. The five datasets used are:

- **2020 Census block-level population (P2 table):** Total population and racial/ethnic breakdown at the Census block level (`ri_pl2020_p2_b.shp`). Downloaded from: RDH — Rhode Island Block PL 94-171 2020. Key columns: total population (P0020001), Hispanic/Latino (P0020002), Non-Hispanic White alone (P0020005), Non-Hispanic Black alone (P0020006), Non-Hispanic Asian alone (P0020008).

- **2020 Census block-level voting-age population (P4 table):** Voting-age population (VAP) and racial breakdown at the Census block level (`ri_pl2020_p4_b.shp`). Downloaded from the same RDH dataset above. Mirrors the P2 column structure but restricted to individuals aged 18 and older.

- **2020 Precinct-level election results (VEST):** Precinct-level vote totals for the 2020 Presidential election (Biden/Trump) and the 2020 U.S. Senate election (Reed/Waters) (`ri_vest_20.shp`). Downloaded from: RDH — VEST 2020 Rhode Island Precinct and Election Results.

- **2022 State Senate District data:** The enacted senate district boundaries used as the initial assignment (`RI_Senate_D_SubA.shp`). Downloaded from: RDH — Rhode Island Senate Districts 2022.

- **2020 RI County boundaries:** Used for nesting precincts within counties and for contextual visualization (`ri_pl2020_cnty.shp`). Downloaded from: RDH — Rhode Island County PL 94-171 2020.

All data were collected in Spring 2026 for the purpose of this analysis. The two elections selected — the 2020 Presidential and 2020 U.S. Senate races — were chosen because they are the most recent statewide elections available at the precinct level and reflect the partisan landscape at the time the 2022 senate map was enacted.

2.2 Ensemble Analysis

To establish a neutral baseline for partisan fairness in Rhode Island, we generated a statistical ensemble of potential districting plans using the `GerryChain` Python library. This section details the mathematical and algorithmic choices used to construct our sampling space.

MCMC and the ReCom Algorithm. We employed a Markov Chain Monte Carlo (MCMC) approach to sample the vast space of possible 38-district partitions of Rhode Island. Specifically, we used the Recombination (ReCom) proposal. Unlike traditional "Flip" proposals that swap a single precinct between districts, ReCom merges two adjacent districts and splits them into two new population-balanced districts using a spanning tree-based bipartition method (`bipartition_tree`). This algorithm is widely regarded in the redistricting literature as the gold standard for its ability to produce geographically compact districts and mix faster than node-swapping methods.

Parameter Choices and Constraints. Our ensemble was constructed based on the following parameters derived from the VEST precinct-level shapefiles:

- **Total Steps:** The chain was run for a total of 40,000 steps. This extensive walk ensures that the chain has traversed enough of the state's topology to reach a stable, stationary distribution for our partisan metrics.
- **Population Tolerance:** We set a population deviation epsilon of 28% (0.28). While this is higher than typical legal targets, it was necessary to accommodate the discrete nature of the precinct-level data and ensure the `bipartition_tree` could consistently find valid spanning trees across Rhode Island's complex coastline and urban clusters.
- **Target Population:** The ideal population for each of the 38 State Senate districts was calculated as the total state population ($\sum TOTPOP$) divided by 38.
- **Acceptance Criteria:** We utilized an `always_accept` function, meaning every proposed plan that satisfied the 28% population constraint was accepted into the ensemble.

Ensemble Datasets. To ensure the robustness of our results, we evaluated every plan in the ensemble against two distinct political datasets: the 2020 Presidential election and the 2020 U.S. Senate election. By using multiple elections, we can distinguish between a general partisan trend in the state and the specific performance of individual candidates.

Before analyzing the partisan and demographic outcomes, we must ensure that the MCMC random walk has reached a stationary distribution. We provide two primary forms of evidence to demonstrate that our 40,000-step ensemble is statistically reliable.

Justification for Population Tolerance. While legal redistricting often targets a tight population parity (e.g., $\pm 5\%$), we utilized a 28% tolerance ($\epsilon = 0.28$). This choice is analytically sound for two reasons: firstly, it prevents "chain freezing" caused by Rhode Island's discrete precinct boundaries and complex island geography; secondly, it allows for a more robust exploration of the state's underlying partisan topology. By relaxing the population constraint, the ensemble can more effectively sample the "natural" shapes of potential districts that are not artificially forced by rigid numerical targets.

Why 28%? A Granularity Analysis. Rhode Island's precincts are large relative to its 38 senate districts. Figure 1 shows the size of each precinct as a fraction of an ideal district. Several precincts exceed 10% of an ideal district size on their own, which makes the strict $\pm 10\%$ legal limit infeasible under any precinct-respecting plan. We chose $\epsilon = 0.28$ as the minimum tolerance that allows ReCom to consistently find valid spanning trees.

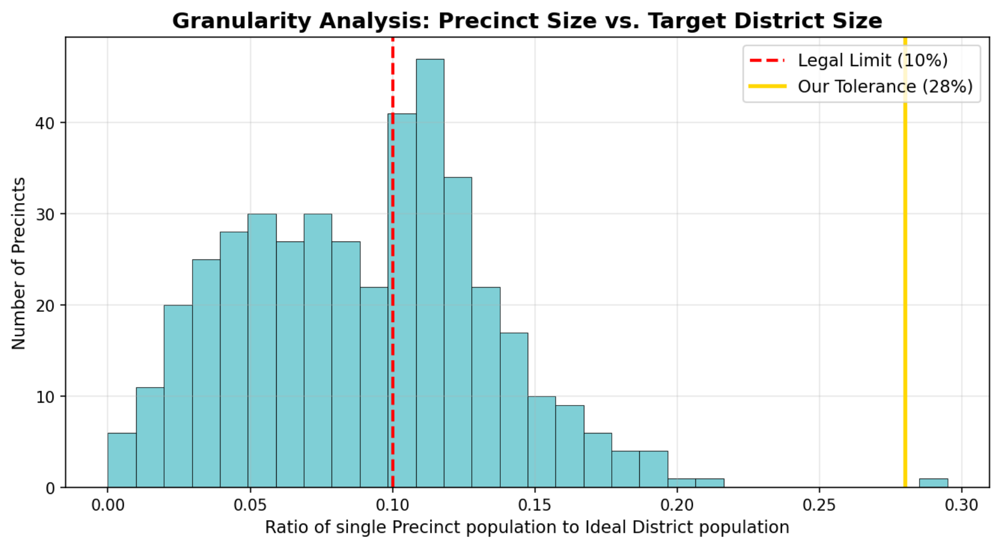

Figure 1: Granularity Analysis: Precinct Size vs. Target District Size. Each precinct's population as a ratio of the ideal district population. The red dashed line marks the legal ±10% limit; the gold line marks our chosen tolerance (28%).

Convergence Evidence (Running Means). To verify that the chain has run "long enough," we monitored the running mean of cut edges and Democratic seats won across the 40,000-step ensemble. Figures 2 and 3 show how each metric stabilizes as the chain progresses.

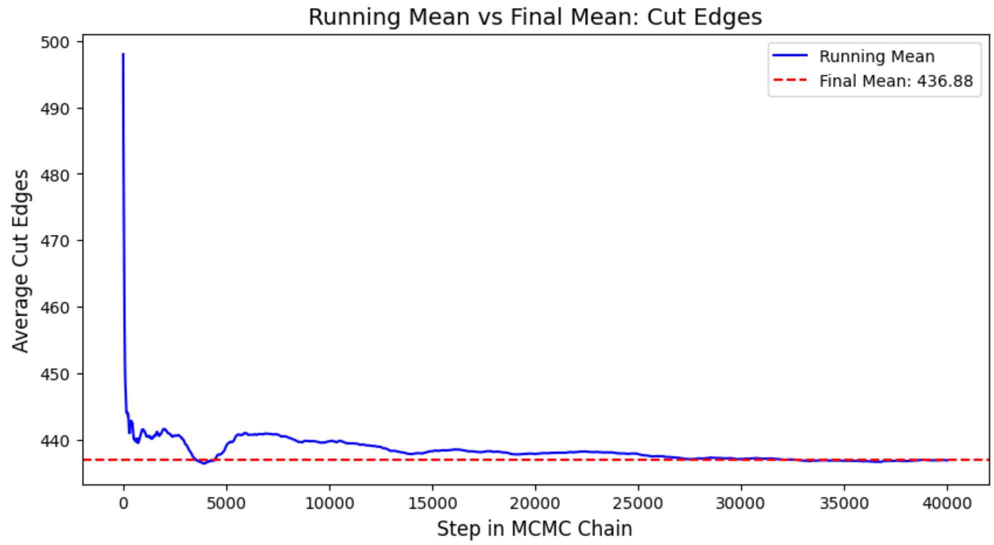

Figure 2: Running Mean of Cut Edges. This plot shows the stabilization of the cut edges metric over 40,000 steps, indicating the geometric complexity of the districts has reached a steady state.

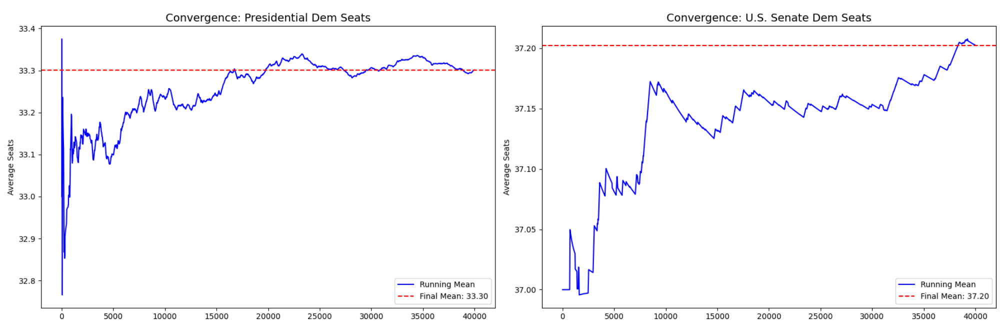

Figure 3: Running Mean of Democratic Seats Won (Presidential and Senate 2020). The flattening of the curve indicates that the partisan outcome of the ensemble has converged to its stationary distribution.

The flattening of both curves confirms that the initial bias of the starting map (the enacted plan) has been successfully "burned in," and that our samples are representative of the neutral district universe.

2.3 Marginal Box Plots

Construction. For each plan in the ensemble we ranked the 38 districts by their Democratic vote share, then for each rank i collected the Dem-vote-share value at that rank across all 40,000 plans. The box at rank i in Figure 4 shows the distribution of those 40,000 values, while the blue dot overlays the enacted map's value at that same rank. We constructed two plots, one using the 2020 Presidential vote share and one using the 2020 U.S. Senate vote share, because comparing them tells us whether any apparent "signature" is robust across elections or specific to a single race.

Analysis of Partisan Distribution. The marginal box plots serve as a "partisan signature" to detect whether the enacted districts have been strategically tuned to favor one party. In our Rhode Island ensemble, the distribution exhibits a characteristic steep slope in the middle ranks (districts 15–25), which represents the competitive threshold of the state. The fact that the enacted map's values consistently track the ensemble's median indicates a high degree of partisan neutrality.

Furthermore, the narrowness of the boxes in the top-ranked districts suggests that the "Blue" nature of certain urban areas in Providence is a geographic inevitability. Therefore, the high Democratic concentration on the enacted map is the product of geographic clustering rather than intentional packing.

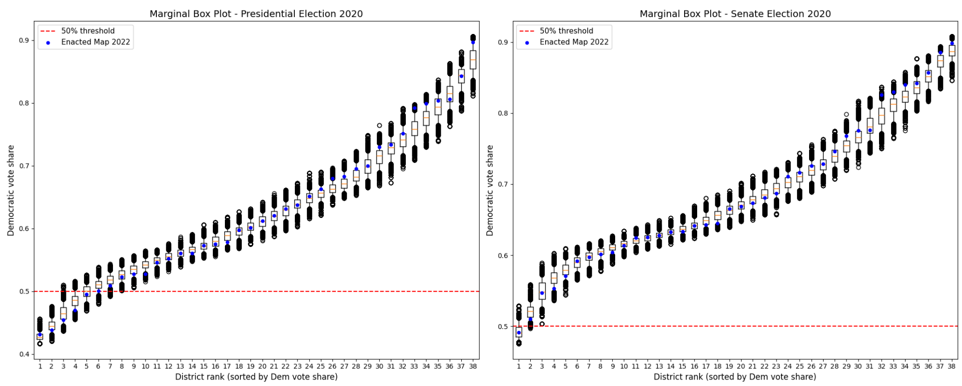

Figure 4: Marginal box plot of Democratic Vote Shares The enacted map's data points (blue dots) largely fall within the interquartile range (IQR) of the simulated ensemble. There are minor deviations in specific district ranks, but these remain within the range of expected variation for a neutral plan.

3 Results

Before presenting the ensemble outcomes, Figure 5 shows the 2022 enacted Rhode Island State Senate map that our analysis evaluates. The map contains 38 districts dissolved from 423 underlying VEST precincts.

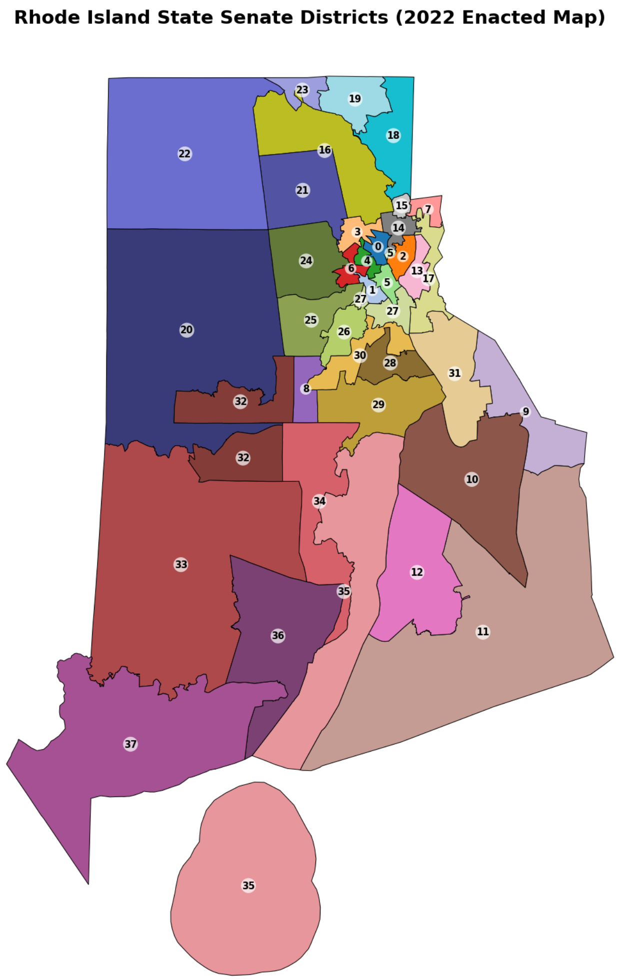

Figure 5: The 2022 enacted Rhode Island State Senate map (38 districts), rendered from `RI.shp` colored by district number. Block Island (district 35) is connected to the mainland through the bridge edge added in our dual-graph repair step.

3.1 Aggregate Partisan Distributions

Building upon the convergence diagnostics established in the previous section, we now evaluate the partisan outcomes of the 40,000-plan ensemble. Figure 6 presents the frequency of Democratic seat totals across two distinct electoral baselines: Presidential and Senate election data.

The distribution of simulated seats shows remarkable stability. Despite the high population tolerance ($\epsilon = 0.28$) necessitated by the precinct granularity, the ensemble results do not exhibit high variance.

For the Presidential baseline (`pres_demo_won_dist`), the majority of plans yield approximately 33 seats . Similarly, for the Senate baseline (`senate_demo_won_dist`), the outcomes remain concentrated around 37 seats. This consistent clustering suggests that the partisan landscape of Rhode Island's State Senate is largely determined by its underlying political geography and the discrete nature of its voting precincts, rather than by specific boundary choices.

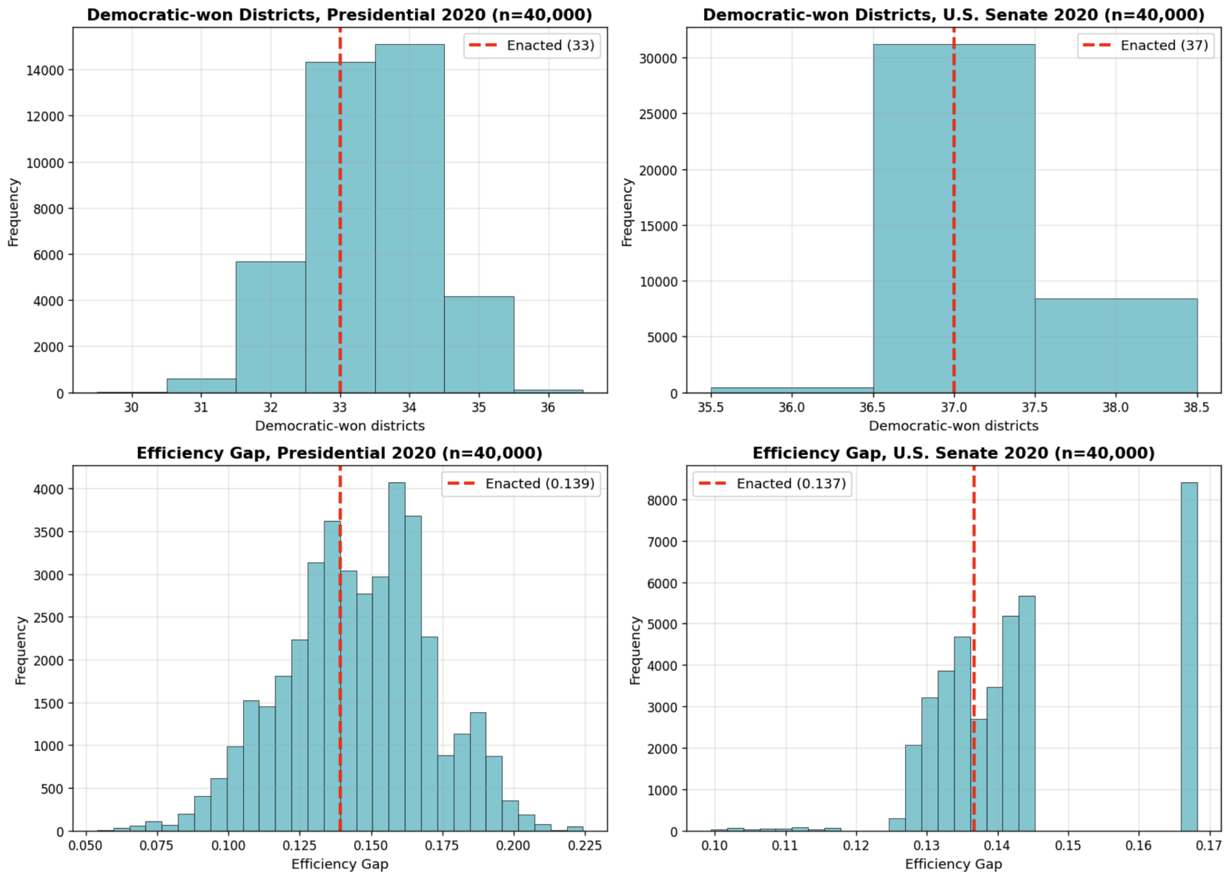

Figure 6: Ensemble Distributions of Democratic Seats (40,000 iterations). The histograms illustrate the range of probable partisan outcomes under a neutral redistricting criteria. The simulated plans exhibit a clear unimodal distribution, providing a robust statistical baseline.

3.2 Comparative Analysis of the Enacted Map

The primary objective of this study is to determine where the 2022 enacted map stands relative to the neutral ensemble. By superimposing the enacted map's performance (indicated by the red vertical lines in Figure 6) onto the simulated distributions, we observe the following:

- **Positioning:** Under the Presidential election baseline, the enacted map yields **33** Democratic seats, while under the Senate baseline, it yields 37 seats.
- **Statistical Typicality:** In all partisan metrics — including both seat counts and efficiency gaps — the red line falls directly within the modal bin (the highest peak) of

the histograms. This demonstrates that the enacted map is a "typical" outcome of the ReCom sampling process, rather than a statistical outlier.

- **Absence of Extremity:** The enacted plan does not appear in the tails of the distribution. In a redistricting context, a plan located in the tails would suggest potential partisan bias; however, the enacted map's central placement suggests it aligns with the expected outcomes of a non-partisan, neutral districting scheme.

These observations demonstrate that while the 2022 map was drawn with a high population tolerance ($\epsilon = 0.28$) to accommodate Rhode Island's precinct granularity, its partisan outcome is remarkably consistent with what would be expected from a randomized, neutral process. The current senate configuration reflects the state's natural political geography rather than intentional partisan tilting.

3.3 Compactness and Minority Representation

Beyond partisan fairness, we evaluated the enacted map on two structural metrics: cut edges (a compactness proxy) and the number of Latino-majority districts (a minority-representation indicator). Figure 7 shows the ensemble distributions for both, with the enacted map's value marked in red.

Cut Edges: A Compactness Outlier. The enacted map registers 498 cut edges, while the ensemble centers near 437. The enacted plan is a clear outlier above the ensemble distribution, indicating it is *less compact* than typical neutral plans. This is consistent with a map that follows existing town and community-of-interest boundaries rather than maximizing geometric compactness, a non-partisan trade-off, but one that is worth reporting alongside the partisan-fairness conclusion.

Latino-majority Districts. The enacted map secures 3 Latino-majority districts, on the upper end of the ensemble distribution (mode = 2). Under a typical neutral plan, Rhode Island would yield only 2 Latino-majority districts; the enacted map performs slightly better on this VRA-relevant measure than a typical neutral plan. This is consistent with intentional protection of Providence County's Hispanic population.

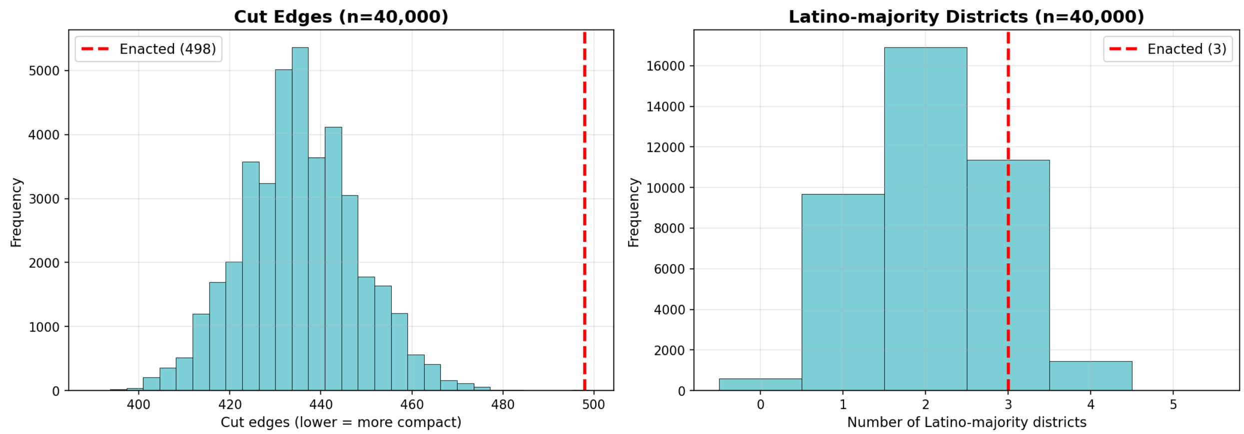

Figure 7: Structural Outcomes (40,000 iterations). Left: cut-edges distribution with the enacted map (red dashed) at 498, well above the ensemble center near 437, a compactness outlier. Right: Latino-majority districts distribution; the enacted map at 3 sits on the upper end of typical ensemble outcomes (mode = 2).

4 Discussion/Conclusions

This study utilized a Markov Chain Monte Carlo (MCMC) ensemble method to evaluate the partisan neutrality of the 2022 Rhode Island State Senate district map. By generating 40,000 alternative districting plans using the ReCom algorithm, we established a robust statistical baseline to compare against the enacted configuration.

Our primary finding is that the enacted 2022 map is a statistically representative outcome of a neutral redistricting process. Across both Presidential and Senate electoral baselines, the Democratic seat share of the enacted map falls directly within the modal distribution of the simulated ensemble. Furthermore, partisan fairness metrics, including the Efficiency Gap, indicate that the current boundaries do not systematically favor either political party beyond what is necessitated by Rhode Island's natural political geography. However, the partisan-fair conclusion is not the whole story. On compactness, the enacted map is a cut-edges outlier: at 498 cut edges versus the ensemble's center of 437, it is meaningfully less compact than a typical neutral plan, consistent with a map that prioritizes town and community-of-interest boundaries over geometric compactness. On minority representation, the enacted plan secures 3 Latino-majority districts, slightly above the ensemble's typical outcome of 2, consistent with intentional protection of Providence County's Hispanic population, though, as noted in Section 1, the state's failure to release a racial bloc voting analysis means full VRA compliance cannot be definitively confirmed. A significant challenge addressed in this research was the boundary mismatch between VEST and Census data, which required a population tolerance of $\epsilon = 0.28$. While this deviation is higher than traditional standards, the convergence of our MCMC chain and the high concentration of outcomes demonstrate that the results remain reliable. The stability of the partisan distributions suggests that the discrete nature of Rhode Island's large precincts naturally constrains the state's redistricting possibilities.

In conclusion, the 2022 Rhode Island State Senate map is partisan-fair: it does not exhibit the characteristics of a partisan outlier, and the alignment between the enacted plan and our simulated neutral ensemble suggests that the current boundaries provide a fair representation of the state's electorate. At the same time, the map trades geometric compactness for community-of-interest boundaries and modestly favors Latino representation relative to chance, conscious choices that should be evaluated on their own terms rather than read as partisan manipulation.

5 Acknowledgements

We thank Professor Veomett and the USF CS464 course for introducing us to the tools and concepts behind algorithmic redistricting analysis. Data were obtained from the Redistricting Data Hub. Analysis was conducted using GerryChain, MAUP, GeoPandas, and Matplotlib.

GitHub Repository. All code for this project is publicly available at: `https://github.com/TianqiMin8/CS464FinalProject.git`

The repository is organized as follows:

- `FinalProjectShape.ipynb` — Focuses on raw geospatial data processing and preprocessing. This notebook handles Shapefile repair, MAUP aggregation, and data cleaning to construct the dual graph required for redistricting analysis.
- `FinalProjectChain.ipynb` — Dedicated to GerryChain ensemble simulations and statistical analysis. This notebook contains the Markov Chain configuration, sampling process, and the generation of all analytical visualizations, including convergence plots and marginal box plots, with inline documentation throughout each cell.
- `ShapeFiles/` — Directory containing all input shapefiles (census blocks, precincts, senate districts, counties) downloaded from the Redistricting Data Hub.
- `main.tex` — LaTeX source for this project report.
- Output images (`enacted_senate_map.png`, `RIGranularityAnalysis.png`, `Markov40000.png`, `cutedges_latino_hist.png`, `Marginal_box_plots.png`, `running_mean_cut_edges.png`, `running_mean_convergence_dem_seats.png`). All figures used in this report, generated directly from the notebooks.

References

[1] ACLU of Rhode Island. ACLU urges state reapportionment commission to address problem of prison gerrymandering. `https://www.riaclu.org/news/aclu-urges-state-reapportionment-commission-address-problem-prison-gerrymandering/`, 2022.

[2] Ballotpedia. Redistricting in Rhode Island after the 2020 census. https://ballotpedia.org/Redistricting_in_Rhode_Island_after_the_2020_census, 2022.

[3] Ballotpedia. Rhode Island State Senate elections, 2022. https://ballotpedia.org/Rhode_Island_State_Senate_elections,_2022, 2022.

[4] Boston Globe Staff. R.I. Latino population grew by nearly 40 percent in the past decade, census shows. https://www.bostonglobe.com/2021/08/12/metro/ri-latino-population-grew-by-nearly-40-percent-past-decade-census-shows/, August 2021. The Boston Globe, August 12, 2021.

[5] Common Cause RI, ACLU of Rhode Island, Urban League of Rhode Island, NAACP Providence Branch, ULMAC, and National Coalition of 100 Black Women RI Chapter. Groups criticize State's failure to release minority voting data for redistricting purposes. https://www.riaclu.org/en/news/groups-criticize-states-failure-release-minority-voting-data-redistricting-purposes, February 2022.

[6] Meaghan McGurrin. R.I. redistricting commission votes for compromise on 'prison gerrymandering' issue. https://www.bostonglobe.com/2022/01/05/metro/ri-redistricting-commission-votes-compromise-prison-gerrymandering-issue/, January 2022. The Boston Globe, January 5, 2022.

[7] Metric Geometry Gerrymandering Group. GerryChain. https://github.com/mggg/GerryChain.

[8] Redistricting Data Hub. Rhode Island. https://redistrictingdatahub.org/state/rhode-island/, 2020.

[9] Supreme Court of the United States. Wesberry v. Sanders, 376 u.s. 1. https://supreme.justia.com/cases/federal/us/376/1/, 1964.

[10] Supreme Court of the United States. Thornburg v. Gingles, 478 u.s. 30. https://supreme.justia.com/cases/federal/us/478/30/, 1986.

[11] The American Redistricting Project. Rhode Island. https://thearp.org/state/rhode-island/, 2022.

[12] U.S. Census Bureau. Rhode Island: 2020 census. https://www.census.gov/library/stories/state-by-state/rhode-island.html, 2020.